

Prompt Comparison

Objective

The objective of this exercise is to understand how improving a prompt step by step leads to better AI-generated responses. A clear and detailed prompt helps the AI provide more accurate, structured, and relevant outputs.

Introduction

Prompt Engineering is the practice of writing effective instructions for AI models. Small improvements in a prompt — such as adding context, assigning a role, or specifying the output format — can significantly improve the quality of the response.

In this document, one basic prompt is improved through four iterations and the outputs are compared.

Task

Topic: Explain Machine Learning

Version 1 — Basic Prompt

Prompt

Explain Machine Learning.

Sample Output

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions without being explicitly programmed.

Observation

- Very general
- Short response
- No examples
- No structure

Version 2 — Improved Prompt

Prompt

Explain Machine Learning in simple language.

Sample Output

Machine Learning is a branch of AI where computers learn from data instead of following fixed instructions. It is used in applications such as spam detection, movie recommendations, and voice assistants.

Improvements

- Easier to understand
- Includes practical examples

- Suitable for beginners

Version 3 — More Detailed Prompt

Prompt

Explain Machine Learning for MCA first-year students. Include: Definition, Types, Advantages, Disadvantages, and One real-world example.

Sample Output

Machine Learning is a field of AI that enables computers to learn from data.

Types

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Advantages

- Automation
- Better predictions
- Faster decision-making

Disadvantages

- Requires quality data
- High computational cost

Example

Netflix recommends movies based on user preferences.

Improvements

- Audience specified
- Structured response
- Covers important concepts

Version 4 — Final Optimized Prompt

Prompt

You are an experienced Machine Learning professor. Explain Machine Learning for MCA students. Include: Definition, Types, Advantages, Disadvantages, Real-world example, Comparison table, Conclusion. Use simple language and keep the response within 500 words.

Sample Output

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data without explicit programming.

Type	Description
Supervised	Uses labeled data
Unsupervised	Uses unlabeled data
Reinforcement	Learns through rewards and penalties

Conclusion: Machine Learning is widely used in healthcare, finance, education, and business to automate tasks and improve decision-making.

Improvements

- Assigned an expert role
- Added audience
- Defined output structure
- Included a table
- Added word limit
- Produced a detailed and organized response

Prompt Comparison Table

Version	Prompt Quality	Output Quality	Improvement
Version 1	Low	Basic	Generic response
Version 2	Medium	Better	Simpler language
Version 3	Good	Detailed	Audience and structure added
Version 4	Excellent	High-quality	Role, format, and constraints added

Why Each Revision Produced Better Results

Version 1

The prompt was too broad, so the AI generated only a brief explanation.

Version 2

Adding “simple language” made the explanation easier to understand.

Version 3

Providing the target audience and required sections helped the AI produce a more complete and organized response.

Version 4

Assigning a role, specifying the output format, and setting constraints resulted in the highest-quality response.

Key Learning Points

- Clear prompts improve accuracy.
- Context helps AI understand the task.
- Defining the audience changes the style of the response.
- Specifying the output format improves readability.
- Iterative prompt refinement produces better results.

Screenshot Placeholders

Screenshot 1: Output of Version 1

(Paste screenshot here.)

Screenshot 2: Output of Version 2

(Paste screenshot here.)

Screenshot 3: Output of Version 3

(Paste screenshot here.)

Screenshot 4: Output of Version 4

(Paste screenshot here.)

Conclusion

This exercise demonstrates that prompt quality directly affects AI-generated responses. By refining prompts with additional context, role assignments, formatting instructions, and constraints, users can obtain more accurate, structured, and useful outputs. Prompt Engineering is therefore an essential skill for effectively using Large Language Models in academic and professional tasks.

